BROOKBANKS

1335 Swindon Nev

Technical Note TN42- Old Vicarage Lane S278 Surface Water Design
24th January 2025

1 Executive Summary

1.1 Swindon Eastern Villages Consortium have appointed Brookbanks to produce a surface water drainage design for a residential-led scheme development at South Marston and Rowborough to the east of Swindon, together with areas of mixed-use comprising community; employment; education; local retail uses; and formal and informal open space provision on land east of Swindon and north of A420.

1.2 The scheme is to include an access junction onto Old Vicarage Lane to be adopted under a S278 agreement. The junction will require an approved drainage system to discharge Condition 30– Old Vicarage Lane Access Design (ref. S/OUT/13/1555).

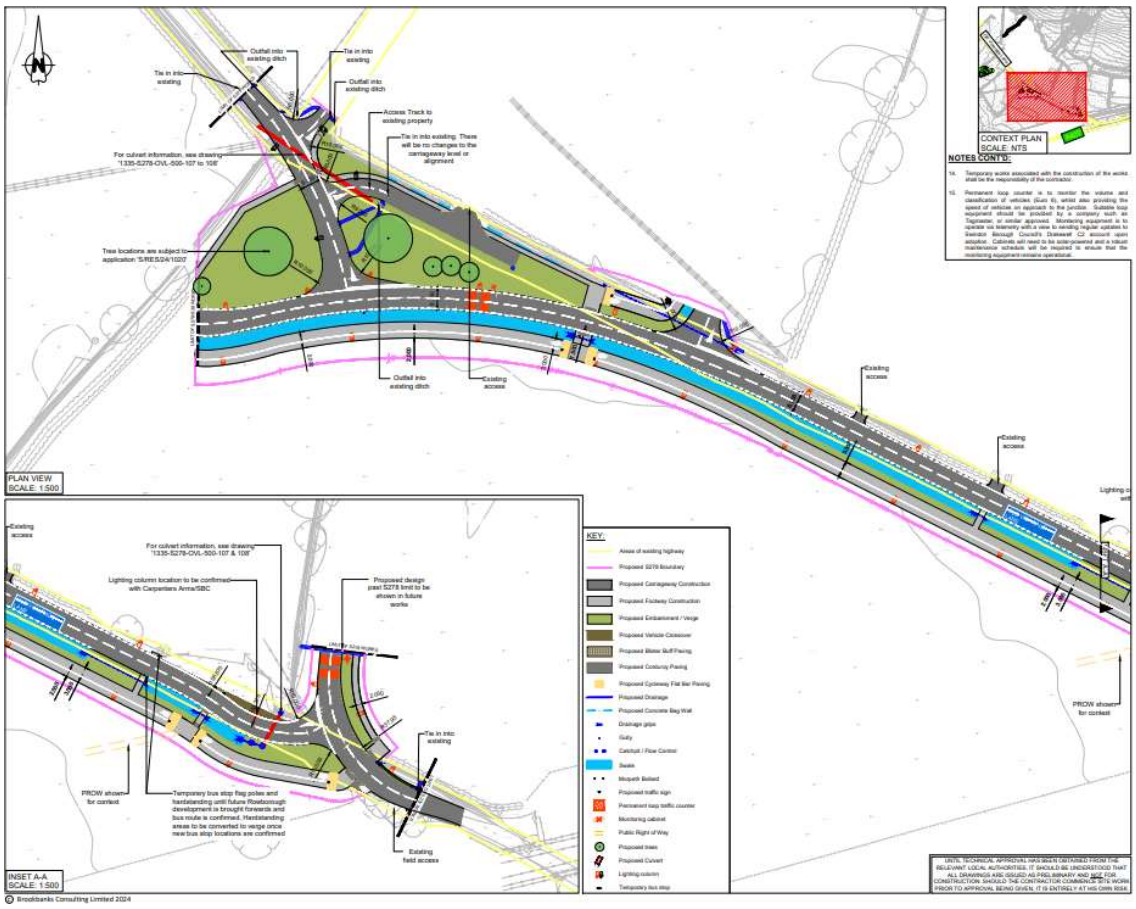
1.3 We believe this can be achieved with an over the edge swale drainage system to attenuate and treat the surface water run off as set out in this note and on drainage plan ref **1335-S278-OVL-500-100 Drainage Layout**. The proposal accords with the approved Flood Risk Assessment 1335 FRA01 Rv3 and the reports prepared to discharge Conditions 40 & 41 relating to site wide drainage and flood mitigation.

2 Proposed Junction

2.1 The proposed layout consists of

- A new priority junction with Old Vicarage Lane (OVL)
- Retained access to existing properties and Nightingale Lane
- New accesses to serve the future phases of the site
- Connection to the proposed primary street to serve the South Marston Phase
- Footways and Cycleway

Shown in the extract below.



Proposed Junction Layout (Extract Brookbanks GA ref 1335-OVL-100-101)

3 Surface Water Run Off

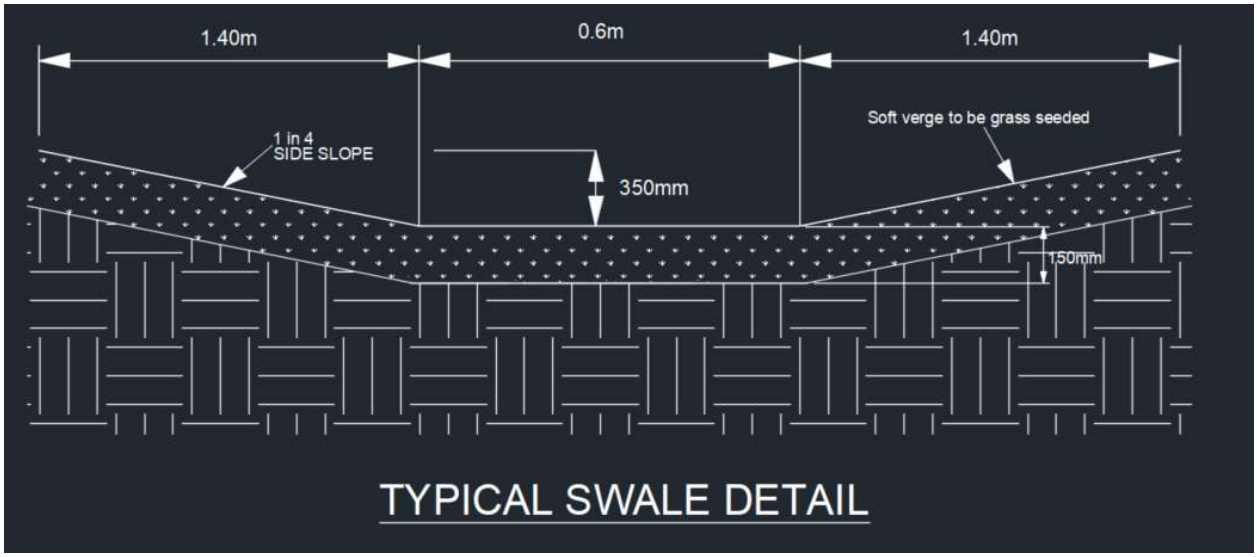
Existing Highway Drainage System

3.1 A topographical survey has been undertaken to verify the extent of the existing highway drainage system draining the OVL. Members of the Brookbanks team have visited site for a walkover and observed the existing highway drainage system to consist of a combination of over the edge drainage and gullies, both of which outfall to ditch courses

3.2 The proposal will be to retain and improve the existing drainage system where the alignment remains unchanged.

Proposed Highway SuDS

3.3 The S278 works will result in an increase of surface water runoff. In line with approved FRA and SWDS a new highway SuDS system is proposed to treat and attenuate runoff as illustrated in **Drawing 1335-S278-OVL-500-100 Drainage Layout**. The system contains over the edge drainage to a swale system located within the verge. The swale is under drained to allow connections where gullies are required. It will be up to 4m in width with 1 in 5 side slopes and will be a planted with suitable grass seed and plants.



A typical swale arrangement proposed for the OVL S278 Works

The primary aims of the swale is as follows:

- Final flow and water quality conditioning
- Provide landscaping, amenity and ecological benefits.

Outfall

3.4 Flows will be restricted by a control device and will outfall to a carrier drain crossing OVL and ultimately out falling to the existing ditch to the east. Further work will be undertaken to the existing ditch course in future works associated with Rowborough as set out in the SWDS.

Supporting measurements and calculations

3.5 We have calculated the following using Microdrainage Source Control Software for storm events up to and including a 1 in 100-year event plus 40% climate change.

- Additional Impermeable area approximately **0.5ha**
- Flow restricted to **2.1l/s** to mimic existing flows from the site.
- Volume of storage required approximately **400m^3**

3.6 The proposed highway drainage arrangements, including the incorporation of a Swales, have resulted in no increase to the flow rates to the existing highway system or surrounding drainage network up to and including the 1 in 100-year storm event plus 40% climate change.

4 Maintenance Plan

Highway Drains

4.1 Once constructed and when the adoption process has been completed, Swindon Borough Council will be responsible for maintaining the highway drainage networks and ensuring their continued operation. Consequently drains, manhole and headwalls will be maintainable by the SBC.

SuDS

4.2 The swales are to be maintained by a Swindon Borough Council in accordance with a Maintenance & Management Plan to be approved by the LPA/LLFA. Detailed plans and design drawings indicating features and areas which will be maintained by Swindon Borough Council will be provided in the Maintenance Plan.

4.3 The development layout provides adequate access and working areas to undertake the above maintenance procedures.

4.4 The Council maintenance team will maintain the basin in accordance with best practice based on a typical maintenance regime shown in the tables below:

Maintenance Schedule	Required Action	Typical Frequency
Regular Maintenance	Remove litter (including leaf litter) and debris	Monthly or as required
	Cut grass for spillways and access routes	Monthly during growing season or as required

	Manage other vegetation and remove nuisance plants	Monthly at start, then as required
	Inspect inlets, outlets and overflows for blockages, and clear if required.	Monthly
	Inspect banksides, structures, pipework etc for evidence of physical damage	Monthly or when required
	Inspect inlets and facility surface for silt accumulation. Establish appropriate silt removal frequencies	Monthly for 12 months, then annually or as required
	Inspect inlets and facility surface for silt accumulation, and establish appropriate silt removal frequencies	Six Monthly
	Check any penstocks and other mechanical devices	Annually
	Tidy all dead growth before start of growing season	Annually
	Remove sediment from inlets, outlet and forebay	Annually or as required
	Manage wetland plants in outlet pool- where provided	Annually
Occasional Maintenance	Reseed areas of poor vegetation growth.	As required
	Prune and trim any trees and remove cuttings	Every 2 years or as required
	Remove sediment from inlets, outlet and forebay and main basin	Every 5 years or as required
Remedial Actions	Repair erosion or other damage by re-turfing or reseeding	As required
	Re-alignment of riprap	As required
	Repair/ rehabilitate inlets, outlets and overflows	As required
	Relevel uneven surfaces and reinstate design levels.	As required

5 Flood mitigation measure

Existing Flooding Issues

5.1 Surface Water flooding has been reported by South Marston Parish Council to the south of Old Vicarage Lane close to the Carpenters Arms.

Improving the situation

5.2 The delivery of the entire scheme will provide betterment to the existing surface water run off either through the provision of the new Surface Water network or the overland flow routes. Specific measures are identified in Brookbanks Technical Note 39. We propose that the works identified in the report for the OVL are undertaken as part of the approved S278 work.

5.3 The proposed works include the existing culvert crossing OVL near to the Carpenters Arms being replaced with a new culvert, to extend across the full width of the proposed road. The new culvert will be cleared with the existing ditch outfall cleared. The culvert will be a Swindon Borough Council asset and therefore we will recommend any maintenance to be undertaken by the council's maintenance team.

5.4 The drainage design will improve the existing situation with the inclusion of the roadside swale to capture and convey flows away from highway and the inclusion of a separate outfall from the site to the ditch will reduce flows through the existing culvert.

5.5 Through additional consultation with Richard Bennett, LLFA we have provided further detail and additional measures downstream of the Carpenters Arms and into Rowborough that will be provided as part of the Rowborough site to further alleviate previous flooding issues.

Exceedance Flows

5.6 The wide verge and swale will allow a route for exceedance flows ensuring any flooding in extreme events is diverted away from habitable dwellings. The proposed road also crossfalls towards the swales, away from any dwellings.

6 Next Steps and Recommendations

6.1 The proposed over the edge swale system is designed in accordance with the approved FRA to allow delivery of the proposed S278 access arrangement at Old Vicarage Lane.

6.2 The developer to ensure that the contractor constructs and maintains the drainage system until the system is adopted by Swindon Borough Council

6.3 Flood mitigation works will form part of the S278 Scheme.